

Multi-View Basketball Player and Ball Tracking with 3D Reconstruction

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Abstract. We build and evaluate a multi-view pipeline for basketball player and ball tracking and 3D reconstruction using three fixed, calibrated cameras at the Sanbapolis facility (Trento). Players are tracked with YOLO, the ball with a classical baseline and a pretrained deep model (WASB/HRNet). Player tracking reaches $F1=0.71$. Ball tracking is much harder, and a standard IoU criterion is misleading here since both trackers estimate a center, not the ball’s true, camera-dependent extent. Under a scale-normalized criterion, WASB ($F1=0.08$) outperforms the classical baseline ($F1=0.04$). 3D player reconstruction reaches a mean error of 0.32m. The open challenges are ball detection and re-merging global IDs that fragment when a cross-camera correspondence is lost and regained.

1 Introduction

Multi-camera tracking turns video into structured trajectories for tactical analysis, without wearable sensors. Basketball is hard: fast play, dense occlusion, and a small, often-hidden ball make a single view ambiguous for re-identification and depth. We build a pipeline for a basketball action recorded at Sanbapolis (Trento) with three fixed, calibrated cameras: players tracked with YOLO [6], the ball with two methods, a classical baseline and a pretrained deep model [4]. We evaluate with standard tracking metrics [5] and report a subtle image-rectification bug, present even in the course reference script, that substantially changed our results (Sect. 2.2).

2 Method

2.1 Dataset and Annotation

Our data is a single basketball action captured at Sanbapolis with three fixed, pre-calibrated 4K cameras at 25 fps (~ 21 s, 525 frames each), more views than the two required for a two-person team. Ground truth was annotated in Roboflow (COCO format) at 5 fps, for 309 frames total. Annotators boxed every visible player, referee, and the ball, with players labeled by team color and jersey number (e.g., Red-7) rather than a generic “person” class, and referees separately. For 2D evaluation (Sect. 3.2), player and referee boxes are merged into one “person” class against the ball, since tracking, not re-identification, is our task. Since GT is at 5 fps and tracking at 25 fps, evaluation always compares on matching downsampled frames (Sect. 3.1).

2.2 Camera Model

Each camera is pre-calibrated (provided, not computed by us): pinhole model distortion, intrinsics K , and extrinsics R, t , giving $P = K[R \mid t]$ for triangulation. We undistort detected boxes point-wise (4 corners via `cv2.undistortPoints`) rather than rectifying whole frames, equivalent for localization.

2.3 2D Detection and Tracking

Players. We detect and track players with a pretrained YOLO26n model [6] (COCO “person” class, conf. 0.1) run at 1920×1088, with Ultralytics’ built-in tracker (BoT-SORT [3]) for identity association at the native 25 fps. Detected boxes are rectified as described in Sect. 2.2.

Ball. A single generic detector could not handle the ball’s small size, motion blur, and frequent occlusion, so we iterated through several methods before finding ones that work passably. *YOLO’s COCO “sports ball” class* never fires on this footage (zero detections, all cameras): a basketball at this scale/distance is too far from COCO’s training distribution. A *fine-tuned YOLOv8s*, retrained 150 epochs on a ball-only dataset merging every group’s annotations plus basketball images found online, still failed to generalize reliably. Adding *SAHI* [2] slicing (1500×1500 px tiles, 10% overlap) on top of it, to compensate for the ball’s small footprint in the 4K frame, improved little. Our *classical baseline* (MOG2 background subtraction, shadow removal, morphological cleanup, contour filtering by size/circularity/HSV color) gives a weak but non-trivial signal. *WASB/HRNet* [4], pretrained specifically for basketball, is our best result, though still far from reliable and needing per-camera tiling to fire at all on the widest (full-court) view. We report both trackers in Sect. 3 for comparison.

2.4 Cross-Camera Association and 3D Triangulation

Each camera is tracked on its own, so the same player has a different id in the three views. Before computing any 3D position we must find which track in one camera is which track in another. The rule is simple: two detections of the same player must be consistent with the geometry of the two cameras. For a camera pair and a frame, we try all possible couplings of their tracks, each coupling gives a 3D point, which we project back into the two images, and the further it lands from the original detections, the worse the coupling. We keep the combination with the smallest total error (Hungarian assignment) and give one vote to each coupling whose error stays below 30 px.

Votes are counted over the whole sequence. Two tracks are the same player when they collect at least 5 votes and each is the best match of the other. These links are chained across the three cameras, so that a player seen everywhere ends up with a single global id.

The 3D part is then straightforward. For every frame, a global id seen by at least two cameras gives one 3D point, computed with standard linear triangulation (DLT) [1]. We keep only points inside a box around the court (28×15 m plus a 5 m margin, height from −1 to 3 m): what falls outside comes from a wrong link rather than a real position. The ball is not triangulated: it is tracked in 2D only, extending the procedure to it is left as future work (Sect. 4).

Table 1: Player 2D tracking accuracy ($\text{IoU} \geq 0.5$), per camera and pooled.

Camera	TP	FP	FN	Precision / Recall / F1	MOTP
out2	485	149	277	0.765 / 0.636 / 0.695	0.650
out4	524	31	107	0.944 / 0.830 / 0.884	0.726
out13	894	625	368	0.589 / 0.708 / 0.643	0.669
All	1903	805	752	0.703 / 0.717 / 0.710	0.680

Table 2: Ball 2D tracking accuracy, pooled over all cameras, under both matching criteria.

Tracker	Criterion	TP	FP	FN	Precision / Recall / F1	MOTP / MDE
Classic (MOG2+HSV)	$\text{IoU} \geq 0.3$	1	278	190	0.004 / 0.005 / 0.004	0.437
Classic (MOG2+HSV)	PCK, $\alpha = 0.5$	10	269	181	0.036 / 0.052 / 0.043	56.4 px
WASB/HRNet	$\text{IoU} \geq 0.3$	0	158	191	0.000 / 0.000 / 0.000	n/a
WASB/HRNet	PCK, $\alpha = 0.5$	14	144	177	0.089 / 0.073 / 0.080	49.6 px

3 Results

3.1 Experimental Setup

2D. Predicted and GT boxes are matched per frame/camera/class via Hungarian assignment. We report Precision, Recall, F1, and MOTP (mean IoU over matches) [5]. Players use the standard $\text{IoU} \geq 0.5$ threshold (COCO/PASCAL/CLEAR-MOT). For the ball, IoU is a poor fit: both trackers estimate a center, not the true extent, while GT box size varies sharply by camera (median diagonal 137 to 177 px close-up vs. 54 px full-court), and IoU conflates localization with this scale mismatch. We also report a PCK-style criterion: a hit if the predicted center falls within $\alpha = 0.5$ times the GT box’s diagonal, adapting to local scale instead of penalizing box shape.

3D. Annotated boxes are triangulated into a pseudo-GT 3D point per frame/category (≥ 2 views), matched by Hungarian assignment on 3D distance (threshold 3m). We report coverage, MED, and RMSE, plus trajectory-level continuity (Sect. 3.4), which point accuracy alone does not capture.

3.2 2D Tracking

Table 1 reports player accuracy per camera ($\text{IoU} \geq 0.5$): **out4** (closest, narrowest shot) tracks best, **out13** (widest, most/smaller/occluded people) worst, pooled $\text{F1} = 0.710$ with $\text{MOTP} = 0.680$, i.e. matched boxes overlap well, but many players are missed or spuriously detected (YOLO’s conf. threshold is deliberately low, trading precision for recall).

Table 2 compares the two ball trackers under both criteria. Under IoU, both essentially fail ($\text{F1} \leq 0.004$). Under the scale-normalized criterion, WASB clearly beats the classical baseline ($\text{F1} = 0.080$ vs. 0.043) as expected of a trained model, though recall stays the bottleneck for both. Per-camera (not tabulated for space), WASB does best on **out2** ($\text{F1} = 0.162$) and worst on **out13** ($\text{F1} = 0.049$) despite the smallest mean error there ($\text{MDE} = 21$ px vs. ~ 60 to 67 px elsewhere), and the PCK threshold is itself smaller on **out13** (smaller GT ball), so even accurate detections fail it more often.



Fig. 1: 2D tracking

3.3 3D Reconstruction

Our global IDs are anonymous: they carry no link to the annotated players. So we score the reconstruction geometrically, one frame at a time. Each annotated identity seen in at least two views is triangulated into a pseudo-ground-truth 3D point (1085 points in total, players and referees only). In every frame we pair each GT point with the nearest reconstructed position, one-to-one (Hungarian assignment), and keep the pair only if the two lie within 3 m. Anything unpaired is a miss. The error of a GT point is its distance to that paired position. A row of Table 3 is therefore the average of these per-frame distances for one identity, not the error of a single track followed over time.

560 of the 1085 GT points (51.6%) are matched this way. On them the mean error is 0.321 m and the RMSE 0.531 m. All twelve identities stay under one meter, between 0.144 m (*White_27*) and 0.555 m (*White_34*). The ball is not in the table because we do not triangulate it (Sect. 2.4).

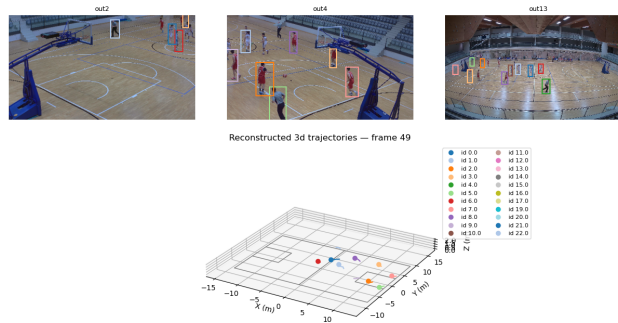


Fig. 2: 3D reconstruction animation frame

3.4 Trajectory Continuity

Point accuracy says nothing about identity continuity over time, which task 3a asks for explicitly. For each of the 22 global 3D tracks, we measure *continuity* as the fraction of frames within its own first-to-last-frame span with a reconstructed position (1.0 = no gaps): 0.929 on average (median span 104 frames = 4.2 s,

Table 3: 3D position error by identity, sorted by mean error

Identity	Mean [m]	<i>n</i>	Identity	Mean [m]	<i>n</i>
White_27	0.144	49	Refree_1	0.319	51
Red_2	0.178	79	Red_9	0.385	28
Refree_2	0.200	14	Red_11	0.395	52
Red_23	0.274	47	White_14	0.410	39
Red_7	0.278	43	White_34	0.555	49
White_16	0.285	65			
White_2	0.288	37			

mean 2.55 gaps/track), i.e. tracks are mostly contiguous once established. The more telling number is the count itself: 22 global IDs against at most 12 distinct annotated identities on court (10 players + 2 referees, Sect. 2.1). Vote-based cross-camera association (Sect. 2.4) has no appearance model, so once a track’s cross-camera correspondence is lost (e.g. occlusion in one view) and later re-acquired, it gets a new global ID rather than merging back, fragmenting some individuals into multiple tracks, which the high mean continuity above does not reveal on its own.

4 Conclusions

We built and evaluated a three-camera pipeline for player and ball tracking with 3D reconstruction, exceeding the two views required for a two-person team. Player tracking reaches F1= 0.710 and 3D reconstruction a mean error of 0.321 m (51.6% coverage), validating the pipeline for the common case. Ball tracking remains the weak point: even WASB, our better tracker, stays far from reliable, and is tracked only in 2D, not yet triangulated. Identity also fragments, 22 global IDs for 12 people on court, because the sequence-wide association vote carries no temporal or appearance-link, consequently the 3D track merging strategy is left as future work.

References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
2. F. C. Akyon, S. O. Altinuc, and A. Temizel, “Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection,” in *Proc. IEEE Int. Conf. on Image Processing (ICIP)*, 2022, pp. 966-970.
3. N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, “BoT-SORT: Robust Associations Multi-Pedestrian Tracking,” arXiv preprint arXiv:2206.14651, 2022.
4. S. Tarashima, M. A. Haq, Y. Wang, and N. Tagawa, “Widely Applicable Strong Baseline for Sports Ball Detection and Tracking,” in *Proc. British Machine Vision Conference (BMVC)*, 2023.
5. K. Bernardin and R. Stiefelhagen, “Evaluating Multiple Object Tracking Performance: The CLEAR MOT Metrics,” *EURASIP J. Image and Video Processing*, vol. 2008, 2008.
6. G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLO,” 2023. [Software] <https://github.com/ultralytics/ultralytics>